

Show, Attend, and Tell – Neural Image Caption Generation with Visual Attention

Aditya Iyer, Aitzaz Sheikh, Joshua George, Tailai Ying

Introduction

- Image captioning sits at the intersection of vision and language: the model must both perceive what is in an image and describe it in fluent natural language.
- Show, Attend and Tell (Xu et al., ICML 2015) applies soft visual attention to captioning. At every generated word, the decoder learns where to look in the image, rather than collapsing the image to a single fixed vector, making the model more accurate.
- We reproduce the soft-attention variant on MS COCO 2014 to verify the paper's central claim and study its attention behavior, laying foundational understanding for modern vision-language models (CLIP, BLIP, ViT-based VLMs).

Methodology

- Used MS COCO (Karpathy splits) for training and evaluation.
- Extracted 14×14×512 VGG-19 feature maps, treating each grid cell as an image region.
- Implemented soft attention to compute word-specific weights over visual regions and used an LSTM decoder to generate captions word-by-word from attended image features.
- Evaluated captions using BLEU-1–4 and METEOR against ground-truth captions.

Sample Captions

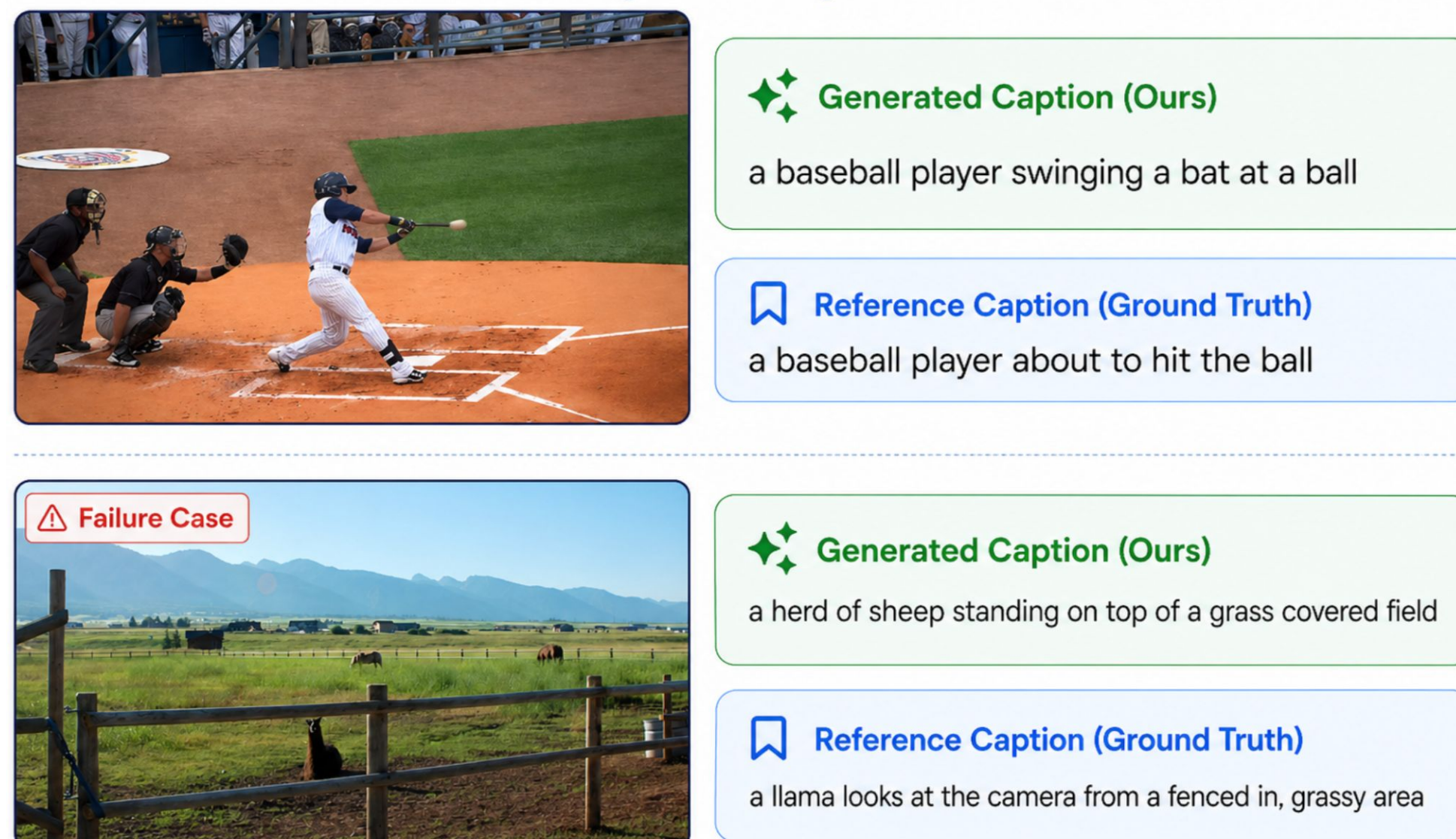


Figure 1: Sample of a successful and failed instance of caption generation.

Model Architecture

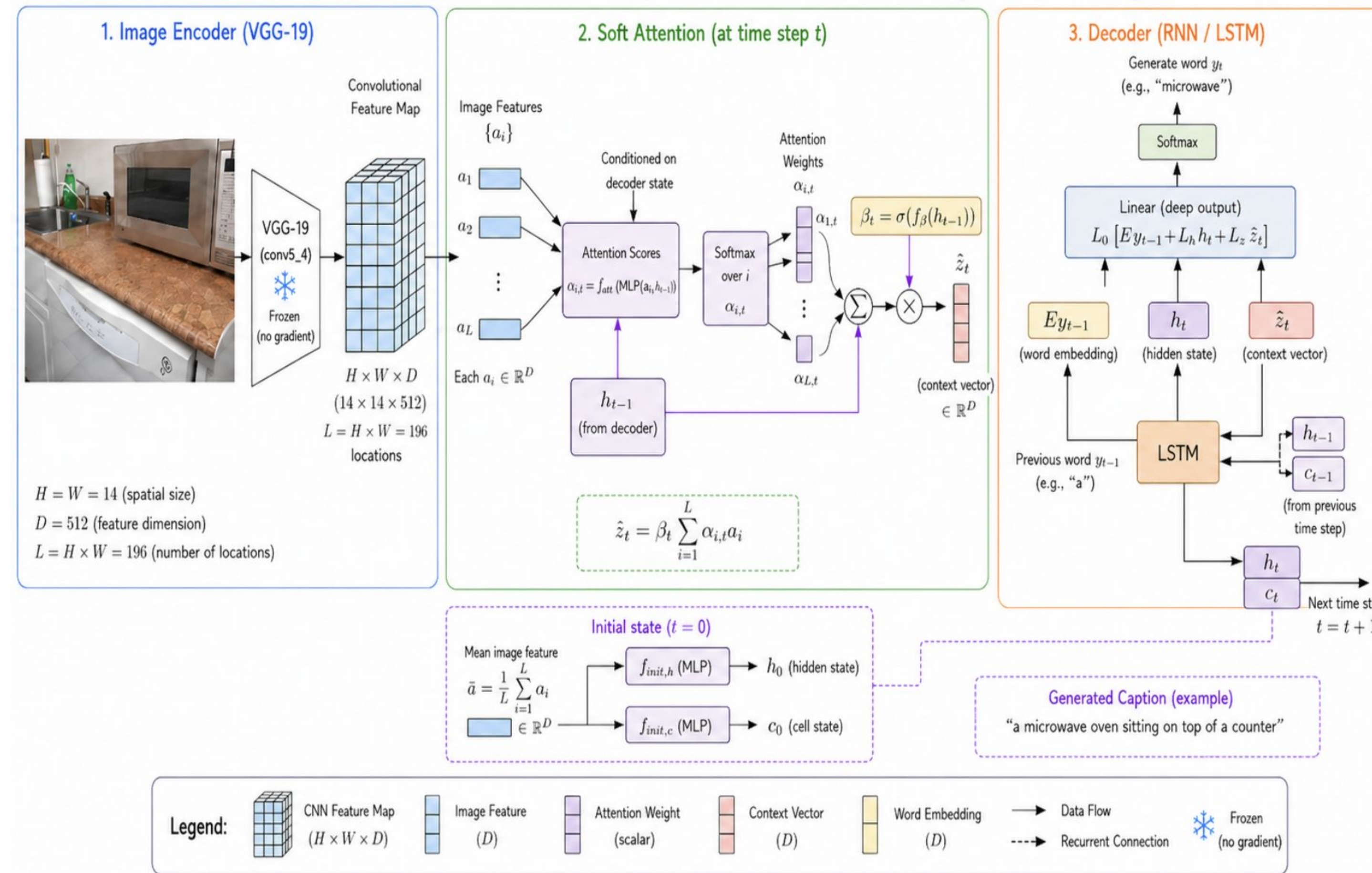


Figure 2: Model Architecture.

Attention Visualization

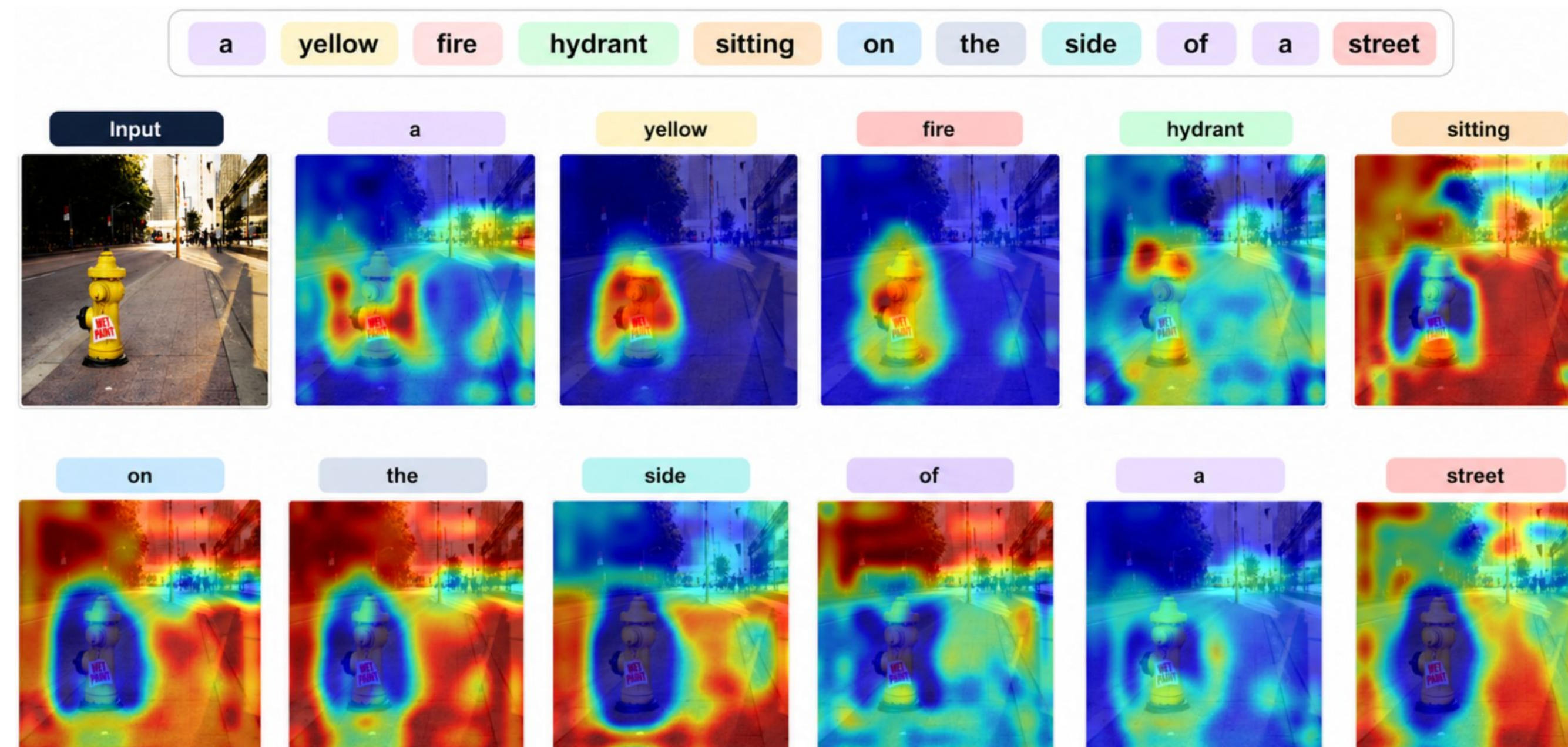


Figure 3: Attention visualization maps per word of the generated caption.

Attention concentrates on the relevant object for content words (nouns, verbs) and disperses for function words (a, the, on), matching the qualitative behavior reported in Xu et al. (2015).

Results

Metric	Paper (Soft Attention)	Our Results
BLEU-1	70.70	63.59
BLEU-2	49.20	45.79
BLEU-3	34.40	32.70
BLEU-4	24.30	23.77
METEOR	23.90	21.55

Figure 4: BLEU and METEOR Evaluation Results.

- We achieve a within 0.6% BLEU-4 and 2.4% METEOR of the paper.
- The gap narrows with n-gram order, suggesting the model matches paper-level fluency but trails on word choice.

Conclusion

We faithfully reproduce Show, Attend and Tell's soft-attention model on MS COCO, landing within 0.6% BLEU-4 of the paper. Generated attention maps localize sharply on objects mentioned in the captions, confirming that learned visual attention improves both caption quality and interpretability.

Future Work

- We can upgrade the encoder to a ViT or CLIP backbone for stronger visual features.
- Instead of letting the model to attend to arbitrary 14x14 grid cells, we can make it attend to meaningful object regions proposed by an object detector.

Reference

Xu, K., Ba, J., Kiros, R., Cho, K., Courville, A., Salakhutdinov, R., Zemel, R., & Bengio, Y. (2015). Show, Attend and Tell: Neural Image Caption Generation with Visual Attention. In Proceedings of the 32nd International Conference on Machine Learning (ICML), 2048–2057.